

Capstone Project: ML-Driven Decision Support for “Aurora Finance”

Scenario:

Aurora Finance is a fictitious mid-size financial services firm that does corporate lending, manages investments, and trades derivatives. They want to leverage ML for better decision-making across the firm:

- Which corporate projects to fund
- How to price loans and assess credit risk
- Where to invest and construct portfolios
- How to monitor and hedge derivative exposures

Students will act as “Data & ML Strategy Consultants”, integrating ML insights into financial decision-making.

Step 0: Project Setup

- Team: 4–5 students (recommended)
 - Timeline: Weeks 1–14
 - Deliverables:
 1. Executive-style project brief
 2. Data exploration & model outputs (notebook or spreadsheet)
 3. Dashboard / visualization summarizing insights
 4. Final presentation/report (10–15 slides)
 - Key objectives: Combine predictive modeling, decision-making metrics, and interpretability.
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Step 1: Corporate Finance Module

Objective: Decide which internal projects Aurora should fund.

Data Provided / Simulated:

- Historical corporate projects: investment amount, expected cash flows, duration, risk rating, department, past ROI
- Market indicators: GDP growth, sector-specific trends

Tasks:

1. Forecast project cash flows using regression / ML (e.g., XGBoost / Random Forest)
2. Predict project success probability / IRR using classification or risk scoring
3. Recommend top projects based on expected economic value (EV) and risk-adjusted return

Deliverables:

- Ranked project list
 - SHAP plots / decision explanation for executives
 - Brief summary of recommendations
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Step 2: Banking Module

Objective: Assess loan portfolio and detect fraud.

Data Provided / Simulated:

- Corporate and retail loan applications: income, balance sheet ratios, past defaults, loan type, duration
- Transactional dataset: anomalies (fraud, late payments)

Tasks:

1. Credit Risk: Predict probability of default (PD) and classify high-risk loans
2. Fraud Detection: Flag anomalous transactions using supervised or unsupervised methods
3. Recommend loan approval or monitoring strategy

Deliverables:

- Risk-scored loan portfolio
 - Fraud alerts / visual dashboard
 - Short summary for lending officers
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Step 3: Financial Markets Module

Objective: Suggest investment strategies for Aurora's portfolio.

Data Provided / Simulated:

- Historical stock prices, macroeconomic indicators, sector indices

- Historical portfolio allocations, returns, risk metrics

Tasks:

1. Predict next-period stock returns using ML (tree-based or regression models)
2. Build optimized portfolio using predicted returns + risk constraints
3. Backtest strategy over last 2 years (simple moving window backtesting)

Deliverables:

- Predicted return table / ranking
 - Recommended portfolio allocation
 - Backtest summary: expected vs realized returns
 - Executive slide: “ML insights for investment decisions”
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Step 4: Derivatives Module

Objective: Hedge portfolio risk and price derivative exposure.

Data Provided / Simulated:

- Option contracts: strikes, maturities, underlying asset prices, historical volatility
- Portfolio from Step 3

Tasks:

1. Predict option prices or implied volatilities using ML
2. Suggest hedging strategy for portfolio (delta/gamma hedging simplified)
3. Evaluate risk exposure metrics (portfolio VaR)

Deliverables:

- Hedged portfolio recommendations
 - Visuals of derivative pricing predictions
 - Risk dashboard
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Step 5: Integrated Executive Decision Dashboard

Combine outputs from all modules into a single, decision-focused dashboard:

- Project funding recommendations

- Loan portfolio & fraud alerts
- Investment portfolio allocation & backtest
- Derivatives risk & hedging recommendations

Deliverables:

- Executive slide deck (~15 slides)
- Decision dashboards (Excel / Tableau / PowerBI / Python interactive)
- Notebook with reproducible ML workflows

Step 6: Executive Summary / Presentation

- 1–2 page summary of all ML-driven recommendations for senior management
- Include key metrics, top 3 action items per function, and risk considerations
- Highlight interpretability & business reasoning, not just ML metrics

Step 7: Evaluation Criteria

Criteria	Weight
ML Model Accuracy & Insights	25%
Business Decision Alignment	25%
Integration Across Modules	20%
Explainability	15%
Executive Communication	15%

📁 Corporate Finance Module — Project Funding

Dataset: corporate_projects.csv

Column	Type	Description
Project_ID	int	Unique identifier

Column	Type	Description
Department	str	Finance, IT, Operations, Marketing
Investment_Cost	float	\$ amount required
Expected_Cashflow_Year1	float	Estimated cash flow year 1
Expected_Cashflow_Year2	float	Estimated cash flow year 2
Expected_Cashflow_Year3	float	Estimated cash flow year 3
Historical_ROI	float	ROI of past projects (similar type)
Market_Growth	float	% industry growth
Project_Risk	str	Low, Medium, High
Success	int	1=successful project, 0=failed (for supervised ML)

Notebook Tasks:

- Regression to predict NPV / cash flows
- Classification for project success
- Ranking projects by expected EV / risk-adjusted return
- SHAP plots for executive interpretation

2 Banking Module — Credit Risk & Fraud

Datasets:

1. loan_portfolio.csv

Column	Type	Description
Loan_ID	int	Unique loan ID
Customer_Type	str	Corporate / Retail
Annual_Income	float	\$
Debt_to_Income	float	Ratio
Credit_History_Length	int	Years
Loan_Amount	float	\$

Column	Type	Description
Interest_Rate	float	%
Loan_Term_Months	int	Duration
Past_Default	int	0/1
PD	float	Target variable for supervised ML

2. transactions.csv

Column	Type	Description
Transaction_ID	int	Unique ID
Customer_ID	int	Link to loan/customer
Amount	float	\$
Transaction_Type	str	Payment, Withdrawal, Transfer
Timestamp	datetime	Time of transaction
Fraud_Flag	int	0/1

Notebook Tasks:

- Predict PD (credit risk) using ML classifiers
- Fraud detection with supervised or anomaly detection methods
- Executive-friendly risk visualization

Financial Markets Module — Investment Strategy

Dataset: market_data.csv

Column	Type	Description
Ticker	str	Stock symbol
Date	date	Daily observation
Open	float	Opening price
Close	float	Closing price

Column	Type	Description
High	float	High price
Low	float	Low price
Volume	int	Trading volume
Sector	str	Industry sector
Macro_GDP_Growth	float	% quarterly GDP growth
Macro_Interest_Rate	float	%
Return	float	Daily return (target)

Notebook Tasks:

- Predict next-day / next-week returns using ML models
- Rank stocks for investment
- Portfolio optimization (Markowitz or simplified)
- Backtesting returns
- Visualizations for executives

4 Derivatives Module — Option Pricing / Hedging

Dataset: options_data.csv

Column	Type	Description
Option_ID	int	Unique identifier
Underlying_Ticker	str	Stock
Strike	float	Strike price
Expiry	date	Expiration date
Option_Type	str	Call / Put
Underlying_Price	float	Current stock price
Historical_Volatility	float	%

Column	Type Description
Interest_Rate	float % risk-free rate
Market_Price	float Observed option price (for supervised ML)

Notebook Tasks:

- Predict option prices using ML regression
- Compare to Black–Scholes theoretical price
- Suggest hedging strategy (simplified delta / gamma)
- Risk dashboard visualization

5 Integrated Executive Dashboard

Notebook Components:

- Combine outputs from all 4 modules into one decision dashboard
- Executive view:
 - Top projects to fund
 - High-risk loans & fraud alerts
 - Recommended portfolio allocation & expected return
 - Derivative hedging recommendations
- Use Plotly / Matplotlib / Seaborn for charts, or Excel/PowerBI for executive-friendly dashboards

6 Notebook Features for Executives

- Pre-filled data exploration cells: summary stats, distributions, correlations
- ML cells with default models: Random Forest, XGBoost, Logistic Regression
- Interpretability plots: SHAP, feature importance
- Guidance comments: explain outputs in business terms
- Downloadable dashboard snapshot: Excel / CSV for executives